

A Novel Curve Fitting Based Discrete Velocity Estimator for High Performance Motion Control

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Abstract—Robotics applications typically need high accuracy and repeatability obtained through precise measurement and control of the robot states which is not available in many cases due to the absence of the necessary sensor motivating the use of estimators. A typical scenario is represented by applications where position sensors (encoders) are present in the robot and velocity measurements are needed by the control scheme. Velocity estimators based on curve fitting (CF) are proposed and evaluated in this paper. A novel CF-based velocity estimator is developed for fitting a smooth curve by computing its polynomial parameters from a given set of position data points. The advantages of this estimator are essentially four: ease of implementation, high operational bandwidth with small phase lag and amplitude attenuation, high computational speed and effectiveness when used with low resolution position sensors. Furthermore, the novel method is model free. The efficacy of the estimator is confirmed by extensive simulation studies while experimental validation evidence is also provided on a single degree of freedom system demonstrating the accuracy and effectiveness of the proposed methods.

I. INTRODUCTION

Traditional robotic applications demand precise motion trajectories requiring accurate sensing and control of motion related states such as position, velocity and acceleration. Some of these states are typically directly measured by sensors installed on the robot joints (e.g. position from joint encoder data) while other states are often computed indirectly from the available measurements (e.g. velocity from position differentiation) or derived from model based estimators. Although sensors such as tachometers and accelerometers are in general limited used mainly due to their higher cost and integration complexity.

With the advent of digital technology many techniques such as e.g. numerical differentiation are used for obtaining these variables from discrete position data, [1, 2]. Numerical differentiation typically returns noisy data particularly when the position resolution is low. It can result in the performance deterioration of control schemes which make use of velocity data obtained by numerical differentiation of the position discrete measurement. The introduced noise caused by the numerical differentiation in the control loop may result in jerky robot motions and low tracking accuracy [3]. A possible solution may be oversampling or the use of low-pass filter, however this substantially limits the velocity estimation bandwidth and may affect the maximum bandwidth

achievable in the controlled system. This motivated the introduction of new techniques for the estimation of these quantities such as velocity/acceleration estimators and observers [4, 5]. A velocity estimator called M/T in [6] provides good velocity estimation for medium to high speed motions, however, since the frequency of the pulses produced by these encoders is proportional to the velocity, the estimation of velocity deteriorates at low speeds due to the low frequency of the generated encoder pulses [5, 7].

To overcome the mentioned problems, extensive work on velocity estimators has been carried out [1, 5], [7, 8]. Traditional estimators as e.g. [9] compute velocity by averaging velocity over a certain sampling interval, however this class of estimators can degrade the performance of the control system and possibly lead to instability due to the lag introduced in the control loop. To cope with these problems, some researchers used Kalman filters [8]. However, the Kalman filter is a model-based approach based on linear dynamic systems discretized in the time domain. Another model free technique usually adopted to reduce the noise of the estimated velocity is the use of low-pass filtering [10, 11]. For instance, the linear differentiator (LD) in [10] is a simple velocity estimator based on a low-pass filter and a conventional differentiator. This method is effective for low velocities but on the other hand it substantially limits the velocity estimation bandwidth therefore affecting the maximum dynamic performance achievable in the controlled system. A similar estimator is the nonlinear tracking-differentiator (TD) introduced by [11] whose implementation is more complex as more parameters need to be set.

The mentioned methods share a similar concept, i.e. a filter combined with a differentiator which implies that they introduce an additional pole in the open/closed loop transfer function, causing magnitude reduction and phase lag. To cope with this, the curve fitting methods such as the least-square CF velocity estimator (LSVE) in [1] can be employed. The key features of these methods are their high operational bandwidth, which introduce small magnitude reduction and phase lag, their model-free nature and the simple implementation. The conventional interpolation techniques can also fit data exactly by passing through all the points [12]. Different from curve fitting interpolation is inappropriate to deal with the data subject to significant errors. Nevertheless, the data feedback by sensors always contain noise and truncation errors. Moreover conventional curve fitting methods such as LSVE require extensive computation to find the parameters by minimizing the square error [13, 14]. This means that such estimators may not be appropriate

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considering the real time control performance which is typically required in feedback control robotic applications.

Motivated by the above, this paper presents a novel CF-based velocity estimator called fast curve fitting velocity estimator (FCFVE). The FCFVE is based on a polynomial fitting method which returns the output signal by multiplying the input signal times a matrix of coefficients instead of calculating the polynomial by minimizing the sum of squared residuals in LSVE. This method is based on a standardized square matrix whose order is the degree of the polynomial. As a result FCFVE improves the computational speed, meaning higher operational bandwidth. Simulation and experimental results confirm that this estimator exhibits higher operational bandwidth and its robustness to input data resolution variations with respect to the conventional estimation methods mentioned above.

This paper is organized as follows. In section II, an overview on the overall problem and LSVE concept is briefly introduced and the design of FCFVE is explained in detail. In section III, a comparative simulation-based analysis is presented to test the properties, efficiency and accuracy of the introduced FCFVE with respect to other methods. Preliminary experimental results are presented in Section IV with Section V addressing the conclusion.

II. CURVE FITTING - BASED VELOCITY ESTIMATORS (CFVE)

A. Overview

1) Overall Problem Description

In motion control of robotic systems the position of the joints which is considered as the input signal $r(t)$ is typically obtained by sensors while velocity observers and estimators [6, 7] can be used to estimate the velocity.

The purpose of curve fitting is to find a function $f(t)$ for the data (t_i, x_i) where $i=0, 1, \dots, n-1$. The function $f(t)$ minimizes the residual which is the difference between the data samples and $f(t)$ under certain criteria. Fig.1 shows the concept of curve fitting.

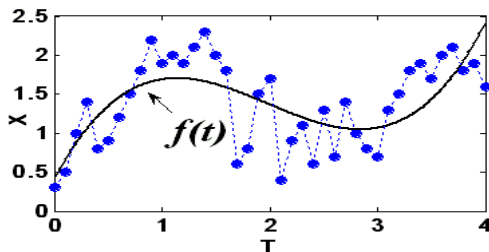


Fig.1 Curve fitting of a set of data points with a polynomial function

This method computes the velocity signal by calculating the derivative of the continuous curve returned by the curve fitting. Different fitting methods such as Least Square (LS), Least Absolute Residual (LAR), or the Bisquare fitting can evaluate the input data to find the parameters. Each method has its own criteria for evaluating the fitting residual in finding the fitted curve. The Least Squares method calculates $f(t)$ by minimizing the square error between the data and the fitting function and is widely used in data fitting [15]. In the next section a concise introduction to least square curve

fitting method is presented as it is then compared with FCFVE presented in this work.

2) Least-Square Curve Fitting Velocity Estimator (LSVE)

The LSVE estimates the signal velocity by means of a least-square fit of the measured discrete position data [1]. The least square fit calculates the best-fitting n - degree polynomial by receiving a set of input data points $\in R^m$, such that $m > n$ [13]. The polynomial of degree n can be written as

$$f(t) = p_0 + p_1 t + p_2 t^2 + p_3 t^3 + \dots + p_n t^n \quad (1)$$

where t is the independent variable. The coefficients of the polynomial in (1) can be calculated by solving a system of m equations [1]. The estimated position at time $t(k) = t_k$ is:

$$\hat{x}_k = p_0 + p_1 t_k + p_2 t_k^2 + p_3 t_k^3 + \dots + p_n t_k^n \quad (2)$$

From (2), the corresponding velocity estimate at time $t(k) = t_k$ can be computed as $d\hat{x}_k/dt$:

$$\frac{d\hat{x}_k}{dt} = p_1 + 2p_2 t_k + 3p_3 t_k^2 + \dots + np_n t_k^{n-1} \quad (3)$$

The LSVE obtains less noisy estimation by minimizing the sum of squared residuals. Generally, the polynomial's degree affects the fitting accuracy, however, the higher the polynomial's degree is the more ill-conditioned the equation set will be [16, 17]. The response of system will also slow down due to the increased computation burden [14]. To cope with this, still guaranteeing the advantages mentioned above, FCFVE is analysed in next section. A third degree polynomial is chosen in this work as a compromise between accuracy and efficiency.

B. Fast Curve Fitting Velocity Estimator (FCFVE)

Although the typically high accuracy of LSCF, it is computationally expensive due to the required computation of the least square fit which ultimately may result in computation delays. This may not be appropriate for feedback control applications. The purpose of the proposed FCFVE is to overcome this major drawback and provide an estimation of the curve fitting with a faster computation speed. In the following lines the FCFVE is introduced.

Let consider the input position signal as:

$$x = f(t) \quad (4)$$

where $f(t)$ represents the continuous expression of the discrete position points after curve fitting and can be expressed by an n -th order polynomial as in (1).

Let now consider one section of the nonlinear curve $f(t)$ in (4), which consists of six discrete points $x_0, x_1, \dots, x_5$, placed at equal time interval $k, k+1, k+2, \dots, k+5$ in one piecewise fitting, Fig. 2.

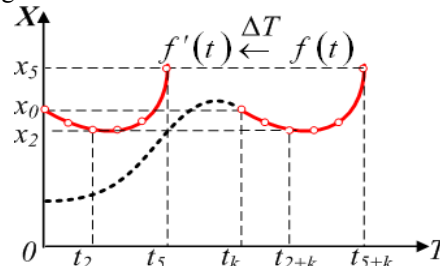


Fig.2 Scheme of the conversion of a nonlinear function

Let now also consider $f'(t) = f(t - \Delta T)$ as the time translation transformation of $f(t)$, Fig. 2. Hence, $f'(t)$ can be written as:

$$f'(t) = x' = x_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots + a_n t^n \quad (5)$$

The procedure so far can be presented schematically in Fig. 2. It is clear that the two red lines have locally the same slope which indicates that the velocity can be also obtained by differentiating $f'(t)$ as follows:

$$v = \frac{df'(t)}{dt} = a_1 + 2a_2 t + 3a_3 t^2 + \dots + na_n t^{n-1} \quad (6)$$

The main objective is therefore to obtain the coefficients in (6). Transposing the zero order term results in the following:

$$\Delta x = x' - x_0 = a_1 t + a_2 t^2 + a_3 t^3 + \dots + a_n t^n \quad (7)$$

Using a third order polyfit it can be written

$$\begin{bmatrix} x_1 - x_0 \\ x_2 - x_0 \\ x_3 - x_0 \end{bmatrix} = \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \end{bmatrix} = \begin{bmatrix} t_1 & t_1^2 & t_1^3 \\ t_2 & t_2^2 & t_2^3 \\ t_3 & t_3^2 & t_3^3 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad (8)$$

From (8) the coefficients can be computed as

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} t_1 & t_1^2 & t_1^3 \\ t_2 & t_2^2 & t_2^3 \\ t_3 & t_3^2 & t_3^3 \end{bmatrix}^{-1} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \end{bmatrix} \quad (9)$$

where $t_1 = \Delta t, t_2 = 2\Delta t, \dots, t_3 = 3\Delta t$.

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} t_1 & t_1^2 & t_1^3 \\ 2t_1 & 4t_1^2 & 8t_1^3 \\ 3t_1 & 9t_1^2 & 27t_1^3 \end{bmatrix}^{-1} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \end{bmatrix} \quad (10)$$

The inverse matrix can be computed as

$$\begin{bmatrix} t_1 & t_1^2 & t_1^3 \\ 2t_1 & 4t_1^2 & 8t_1^3 \\ 3t_1 & 9t_1^2 & 27t_1^3 \end{bmatrix}^{-1} = \begin{bmatrix} t_1 & 0 & 0 \\ 0 & t_1^2 & 0 \\ 0 & 0 & t_1^3 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1^2 & 1^3 \\ 2 & 2^2 & 2^3 \\ 3 & 3^2 & 3^3 \end{bmatrix}^{-1} \quad (11)$$

The constructed matrix $\mathbf{m}_3$ is structured in (11) as

$$\mathbf{m}_3 = \begin{bmatrix} 1 & 1^2 & 1^3 \\ 2 & 2^2 & 2^3 \\ 3 & 3^2 & 3^3 \end{bmatrix}^{-1} \quad (12)$$

The method can be extended to the n^{th} order as $\mathbf{m}_n$. A transition vector can be obtained as

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \mathbf{m}_3 \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \Delta x_3 \end{bmatrix} = \begin{bmatrix} t_1 & 0 & 0 \\ 0 & t_1^2 & 0 \\ 0 & 0 & t_1^3 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad (13)$$

The coefficients vector in (6) can be finally computed as

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} b_1 t_1^{-1} \\ b_2 t_1^{-2} \\ b_3 t_1^{-3} \end{bmatrix} \quad (14)$$

From (14) the parameters of the polynomial in (6) are given. Hence, the derivative value can now be obtained by differentiation in (6). The translation transformation moves the sampled data closer to the origin which can improve the property of curve fitting. The dimension of matrix $\mathbf{m}_n$ will be fixed when the degree of polynomial is determined. In such a calculation, accuracy is determined by precision in the computation of the inverse matrix.

III. SIMULATION RESULTS

The following simulations were performed to examine and compare the performance of the proposed FCFVE with the performance of conventional methods such as LSVE and the traditional linear tracking differentiator LD presented in [10]. The employed input signals $r(t)$ were sine or sawtooth waves with varying frequency. All simulated input position signals $r(t)$ were quantized with the same resolution, i.e. 12 bit/rev to emulate the output signal of the encoder employed in the CompAct™ actuation units developed at IIT, [18-20, 21-23]. All the initial conditions were set to zero. The simulations were performed on an Intel(R) Core(TM) i3 CPU M370 2.40 GHz computer.

The LD can be represented by the following set of equations:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -R^2(x_1 - r) - \beta_2 R x_2 \end{cases} \quad (15)$$

where x_1 is the reference signal as position, x_2 is the derivative signal as velocity whereas β_1, β_2 and R are positive design parameters. The parameters were set as: $R = 25, \beta_1 = 3.0$, and $\beta_2 = 4.0$ for the LD, keeping the same values employed in [10].

The 4th order Runge-Kutta discrete LD method can be formulated as

$$\begin{cases} x_1(k+1) = x_1(k) + \frac{1}{6}(a_1 + 2a_2 + 2a_3 + a_4) \\ x_2(k+1) = x_2(k) + \frac{1}{6}(b_1 + 2b_2 + 2b_3 + b_4) \end{cases} \quad (16)$$

where a_i and b_i are the i -th parameters of Runge-Kutta method and k is an index which refers to the k^{th} sampling instant.

For the FCFVE method which is proposed in this paper a third-order polynomial fitting was used in this simulation study. A third-order polynomial fitting is also utilized for LSVE method to enable a fair comparison.

A. Estimation of velocity from a sinusoidal input position signal

The simulations of LD and LSVE, FCFVE were initially validated by tracking a sinusoidal input reference signal with a frequency of 20Hz and an amplitude of 0.5 rad, $r(t) = 0.5 + 0.5\sin(20 \cdot 2\pi t)$. The frequency of this input simulated signal was selected approximately more than 3 times higher of the natural frequency of the voltage to link velocity transfer function of the actuator in [18] in order to show the behaviour of the different estimators in the high frequency domain.

The results are shown in Fig. 3, showing that the FCFVE outperforms other estimators: higher accuracy and smaller time lag are achieved. However, a time delay in the initial phase can be noticed compared with the actual velocity and this occurs due to the initial number of samples required by the algorithm to compute the first estimated velocity value. Regarding the other methods, the LD exhibits a large lag and small accuracy. Compared to LSVE, FCFVE has a smaller overshoot and faster response.

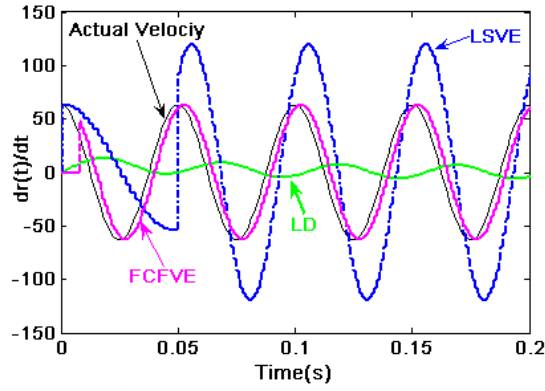


Fig.3 FCFVE performance against LD and LSVE for a 20Hz sinusoidal position profile.

In a second simulation the frequency of the signal was decreased in order to test the performances of the same estimators in the low frequency spectrum. Similarly as in the high frequency case, this scenario also may be critical as demonstrated in [5, 7]. In addition to FCFVE, LD and LSVE methods, the conventional numerical differentiation technique (ND) is compared as well. The simulated input position signal is in this case a sinusoidal signal of amplitude 0.5 rad and frequency of 1Hz. $r(t) = 0.5 + 0.5\sin(1 \cdot 2\pi t)$. The simulation results are shown in Fig.4.

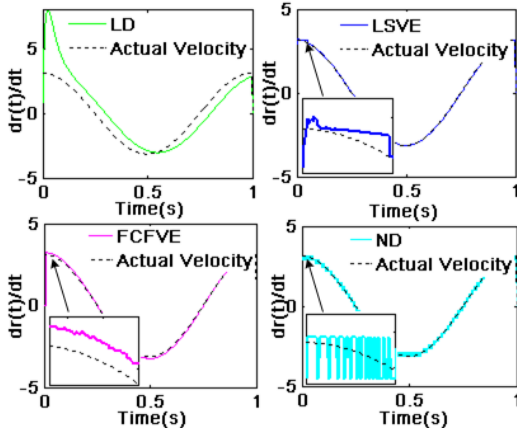


Fig.4 FCFVE performance against LD, LSVE and ND for an 1Hz sinusoidal position profile.

The traditional LD shows larger initial overshoot and lag when compared with the other two curve fitting methods. As it can be noticed in the zoomed area in Fig.4, FCFVE obtained a relatively smoother velocity signal than that computed by LSVE and ND.

B. Tracking sawtooth wave position signal

To compare the effects of velocity estimations while estimating different types of signals, a sawtooth wave was employed as simulated input signal. This signal is mainly employed to evaluate the behaviour of the different methods when sudden and sharp position changes occur. The performances of the different methods in estimating a sawtooth wave signal are shown in Fig.5. The signal is a sawtooth wave with frequency of 5Hz and peak amplitude of 1rad.

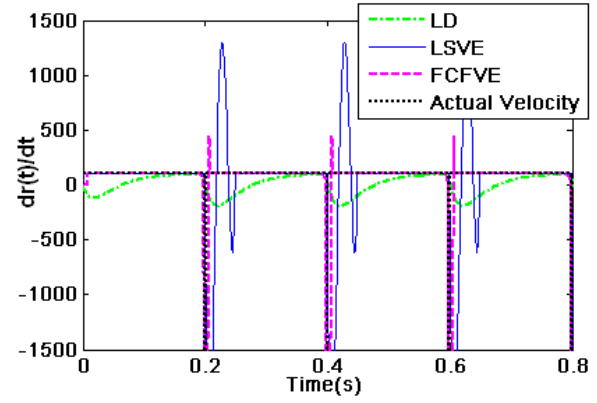


Fig.5 FCFVE performance against LD and LSVE for a 5Hz sawtooth wave position profile.

It is evident that FCFVE provides a good estimation when sharp position variations occur compared with LD and LSVE. LD has slow response which causes inaccurate results while LSVE suffer from time delay and oscillatory behaviour.

C. Quantification of performance

The quantified operation times of different methods in previous simulations are compared in Table I. All the simulations are operated in the same hardware environment using Matlab timer function with the same input signals and parameters as in part A and B. Table I reports the average operation time for one single cycle of estimation computed over the whole simulation time of the simulations reported above. The FCFVE operation time is slightly bigger than that of LD however it outperforms it in most operating conditions as shown above.

TABLE I
OPERATION TIME^a

Input Signals	LSVE	FCFVE	LD
20Hz sin	0.309ms	0.032ms	0.028ms
1Hz sin	7.739ms	0.685ms	0.535ms
5Hz sawtooth	1.186ms	0.148ms	0.133ms

^a ms = millisecond.

In order to characterise the performances of the different methods in the frequency domain, a bode plot of the velocity over the input position signal is presented in Fig.6¹. The input position signal is a chirp of amplitude 1 rad whose frequency increases at a linear rate from 0.1rad/s to 1000rad/s in 50 s. The sampling time is 1 ms. LSVE fidelity is high in the low frequency range. Although LD has a fast computing time it gives poor performance over the frequency spectrum. Differently from the analysed methods, FCFVE shows better performance over frequency. In detail, no phase lag nor attenuation can be noticed below approximately 100 rad/s.

¹ It is worth mentioning that these estimators are nonlinear systems therefore the results reported in this figure may vary for different operating conditions (e.g. due to an input signal amplitude change). However here we report this plot as an indication on the behaviour of the different estimators in frequency.

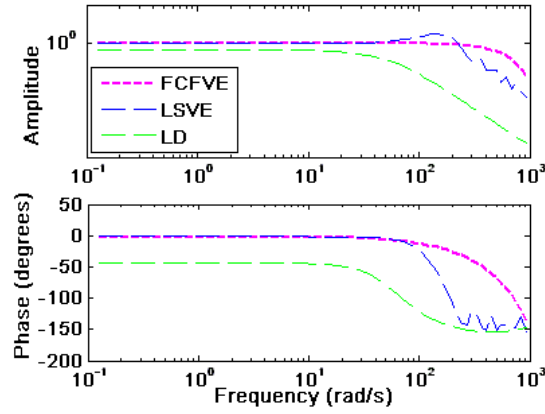


Fig.6 Bode plot of estimators tracking a chirp signal

The analysis of the estimation error was done using the same chirp position signal simulation as that used in Fig.6. The root mean square error (RMSE) which describes the differences between estimated and actual velocities is used to analyse the fidelity of the different methods. Fig.7 shows the RMSE of LD, LSVE and FCFVE varies in frequency. The RMSE values reconfirms that the FCFVE exhibits higher estimation fidelity at any operating frequency compared to the other analysed methods.

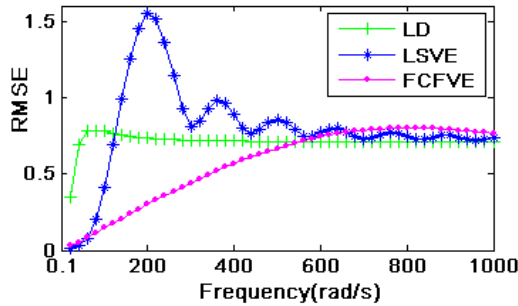


Fig.7 RMSE of the estimation against frequency

The last presented simulation is to analyse the robustness of the different methods to changes in position data resolution. The same input signal employed in Fig.3 is used here.

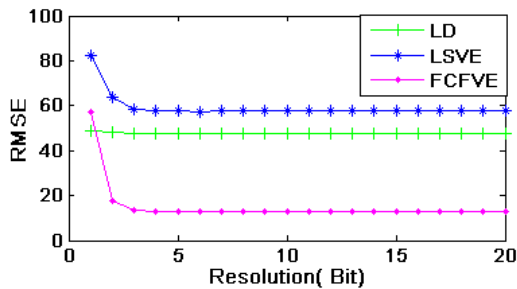


Fig.8 RMSE of the estimation against resolution

The curves in Fig.8 show the RMSE of LD, LSVE and FCFVE with respect to different resolution that varies from 1 to 20 bits. It can be seen that FCFVE shows high fidelity and robustness also when low resolution input data is employed.

IV. EXPERIMENTAL EVALUATION

Experiments were carried out to validate the effectiveness of the proposed estimators in practice. The CompAct™ SEA,

[21], was used as the test platform in the experiment. The actuator in Fig.9 consists of a Kollmorgen RBE1211 frameless brushless motor, coupled to a harmonic drive CSD 17 having a reduction ratio of $N=100$ and a rated peak torque of 40Nm. Two 12bit absolute position sensors have been integrated within the actuation group measuring respectively the angular position of the motor and the deflection angle of the compliant module [21].



Fig.9 The prototype of a compact compliant unit

The feedback motor position signal is measured by an absolute encoder with the resolution of 4096 pulses/rev. The PD controller is used in the position control system as shown in Fig.10.

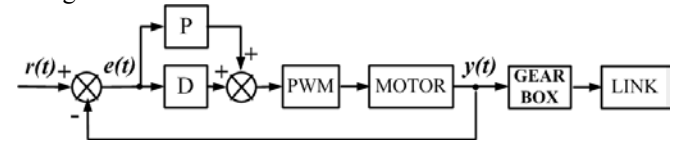


Fig.10 Scheme of PD controller for position control.

The motor reference position signals $r(t)$ were sinusoidal with amplitude of 5 rad and frequency of 2Hz, $r(t) = 5\sin(2 \cdot 2\pi t)$ in this first trial. The method consists in using the feedback position data as input signal for the estimators to calculate the corresponding actual motor velocity. These experiments particularly aimed at evaluating the effect of the estimators as discussed in section 2. The results are shown in Fig.11. The velocities returned by the different estimators are compared with the velocity computed by numerical differentiation to evaluate the effectiveness of the methods.

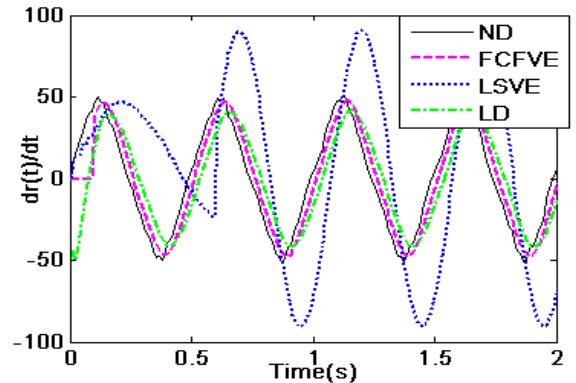


Fig.11 Experimental velocity estimation results of LD, LSVE and FCFVE from a sinusoidal position profile

Fig.11 shows the experimental velocity estimation results for tracking sinusoidal position signal. LSVE is less accurate compared to that obtained with FCFVE and LD. The experimental outcomes confirm the simulation results

presented in simulation previously and demonstrate that the proposed FCFVE provides better velocity estimation than those of other tested methods in terms of both estimation delay and accuracy.

In addition the performance when estimating velocity from a position sawtooth wave is analysed to evaluate the fidelity in estimating velocity when abrupt changes in the position data are given as input to the estimators. The position sawtooth reference sent to the position controlled systems exhibits a period of 0.5 s and peak amplitude of 10 rad. The estimation results are shown in Fig.12.

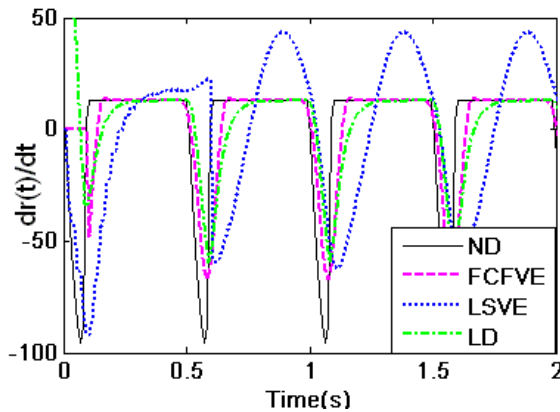


Fig.12 Experimental velocity estimation results of LD, LSVE and FCFVE for a sawtooth wave position profile

It can be noticed that the proposed FCFVE shows a faster response and a better tracking performance than others. From this comparison, it can be seen that the LD has a slower response when the position follows a sawtooth wave while LSVE exhibits overshooting in the estimated output.

V. CONCLUSION AND FUTURE WORK

A fast CFVE was proposed in this paper which can estimate the velocity by using the measured position data. It is a novel curving fitting method which presents the key feature of a relatively high computation speed compared with the traditional curve fitting method like LSVE. In fact, the calculation of the parameters of the estimator is reduced to a simple “multiply and add” operations and can be therefore easily implemented in most DSPs - microprocessors. Another attractive aspect of FCFVE is its model free feature. At same time FCFVE has large operational bandwidth which was depicted by the amplitude and frequency analysis. Both simulations and experiments were carried out to evaluate the performances of the proposed method. The results obtained in the simulations and experiments, confirm FCFVE outperforms conventional linear differentiators and curve fitting methods in the whole frequency spectrum exhibiting smaller time lag, higher accuracy and robustness to variations in position data resolution.

Future work will focus on the implementation of the FCFVE estimator in the viscous damping control of a semi-active friction damper, i.e. the VPDA, [18-20, 23]. In such a device it is expected that the advantages of the FCFVE estimator will be further exploited to improve physical damping feedforward/feedback control.

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